SMARTBRIDGE
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SKILL WALLET
A SmartBridge Product

# Project Initialization and Planning Phase

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| **Date** | June 29, 2026 |
| **Team ID** | |
| **Project Title** | Credit Card Approval Prediction using Machine Learning |
| **Maximum Marks** | 5 Marks |

**Project Proposal (Proposed Solution) Report**

This project automates the credit card approval process using machine learning. It analyses applicant financial and demographic information to predict whether a credit card application should be approved or rejected. The solution reduces manual effort, improves consistency, speeds up decision-making, and minimizes the risk of human error.

| **Project Overview** | |
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| Objective | Develop a machine learning system to predict credit card approval or rejection based on applicant information. |
| Scope | The project includes data preprocessing, model training, evaluation, and Flask-based deployment for real-time credit card approval prediction. |
| **Problem Statement** | |
| Description | Manual credit card approval is time-consuming, inconsistent, and prone to human error. |
| Impact | Improves decision speed, reduces manual effort, and enhances customer satisfaction. |
| **Proposed Solution** | |
| Approach | The dataset is cleaned, preprocessed, and used to train machine learning models. The best-performing model is deployed using Flask for real-time credit card approval prediction. |
| Key Features | • Automated Credit Card Approval Prediction<br>• High Accuracy using Machine Learning<br>• Real-Time Prediction through Web Application<br>• User-Friendly Interface<br>• Fast Decision-Making Process<br>• IBM Watson Machine Learning Deployment Support<br>• Secure and Scalable Architecture |

## Resource Requirements

| Resource Type | Description | Specification/ Allocation |
|---|---|---|
| **Hardware** | | |
| Computing Resources | CPU/GPU Computing | Intel Core i3 or above |
| Memory | RAM | Minimum 4 GB (8 GB Recommended) |
| Storage | Disk Space | Minimum 2 GB Free Space |
| **Software** | | |
| Frameworks | Web Framework | Flask |
| Libraries | ML Libraries | Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn |
| Development Environment | IDE | VS Code |
| **Data** | | |
| Data | Source, Size, Format | Credit Card Approval Dataset (Kaggle), CSV Format, |